Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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Abstract

Brankov, Hansen, and Stevanović (2006) generated 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2026) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques and double star graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 open bounds by their resistance to this mechanism.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
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1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \operatorname{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) \coloneqq \lambda_{\max}\big(L(G)\big) \tag{1}$$

is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic inter-

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est; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] generated 68 conjectured upper bounds for $\mu(G)$ using a backward-reconstruction algorithm that builds expressions evaluating to $2x$ and substitutes d_v or m_v (see [2] for details). These bounds are expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [3] recently applied the cross-entropy method (CEM) of Wagner [4] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [5] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds.

In the present paper we disprove five of the 38 remaining open bounds.

Theorem 1.1 (Main result)**.** *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *at* $t = 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *at* $t = 2m - 1$ (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

(e) *Bound 48 is violated by the path* P_3 (Remark 4.8).

In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [3, 4, 5]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the structural mechanisms underlying all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications and directions for future work.

2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) = D(G) - A(G)$ is positive semidefinite with smallest eigenvalue 0. We write $\mu(G) = \lambda_{\max}(L(G))$ for the *Laplacian spectral radius*.

For each vertex $v \in V$ with $d_v \geq 1$, the *average neighbor degree* is

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [6].

Definition 2.1 (Equitable partition)**.** A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \leq i, j \leq k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([6, Theorem 9.3.3])**.** *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \geq \lambda_{\max}(Q_L),$$

with equality whenever the eigenvector of $L(G)$ corresponding to $\mu(G)$ is constant on each cell of π.

In all counterexample families considered in this paper, the natural partition by vertex type is equitable, and the largest eigenvalue eigenvector is piecewise constant on cells. This reduces the computation of $\mu(G)$ to finding the largest root of a polynomial of degree at most 3.

2.3. The BHS bounds

We now state the five conjectured bounds from [2] that we disprove. All five are listed as open in [3, Table 1]. Bounds 11 and 13 are vertex-type bounds; Bounds 40, 45, and 48 are edge-type bounds.

Conjecture 2.3 (BHS Bound 11 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^3}{d_v^2}.$$

Conjecture 2.4 (BHS Bound 13 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p / d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

3. Counterexample graph families

Definition 3.1 (Star-of-Cliques)**.** For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star)**.** For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars)**.** For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k + 1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu\big(\mathrm{SoC}(K_m, t)\big) = \frac{(m + t + 1) + \sqrt{(t - m)^2 + 2(t + m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m + t + 1)\lambda + (tm + 1) = 0,$$

giving the largest eigenvalue as (3). Since the corresponding eigenvector is piecewise constant on the cells, this is also the largest eigenvalue of $L(G)$. $\square$

Theorem 4.2. *BHS Bound 11,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}, \tag{5}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *at* $t = 2m - 1$*. In particular,* $\mathrm{SoC}(K_8, 15)$ *with* $n = 121$ *satisfies* $\mu(G) \approx 16.796$ *while the bound gives* ≈ 16.000*.*

Proof. By Proposition 4.1, we have $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. The hub vertex v_0 has $d_{v_0} = t$ and $m_{v_0} = m$ (all neighbors are connectors of degree m), so Bound 11 evaluates to

$$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}.$$

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2 + 4m - 4}\big)/2 > 2m$ since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$ (obtained by squaring). The violation persists in a neighborhood of $t = 2m - 1$ whose width grows with m (computations show windows $[5, 7]$ for $m = 4$ up to $[22, 137]$ for $m = 12$), but for fixed m the connector bound $2m_v^3/m^2$ eventually grows as $\Theta(t^3/m^2)$, exceeding the linearly growing μ, so the violation window is finite. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m - 1$*. In particular,* $\mathrm{SoC}(K_9, 17)$ *with* $n = 154$ *satisfies* $\mu(G) \approx 18.815$ *while the bound gives* ≈ 18.000*.*

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^4/m^3 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu\big(\mathrm{SoC}(K_5, 9)\big) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu\big(\mathrm{DS}(k,k)\big) = \frac{(k+3)+\sqrt{k^2+6k+1}}{2}. \tag{7}$$

Proof. The $\mathbb{Z}_2$ symmetry swapping the two hubs decomposes eigenspaces into symmetric and antisymmetric sectors.

Symmetric sector. Quotient matrix $Q_+ = \left(\begin{smallmatrix} k & -k \\ -1 & 1 \end{smallmatrix}\right)$, eigenvalues 0 and $k+1$.

Antisymmetric sector. Quotient matrix $Q_- = \left(\begin{smallmatrix} k+2 & -k \\ -1 & 1 \end{smallmatrix}\right)$, characteristic polynomial $\lambda^2-(k+3)\lambda+2=0$, eigenvalues $(k+3\pm\sqrt{k^2+6k+1})/2$.

Pendant leaves contribute eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, the antisymmetric λ_+ is the global maximum. □

Theorem 4.5. *BHS Bound 45,*

$$\mu(G) \leq \max_{(i,j)\in E}\left(2+\sqrt{(d_i-d_j)^2+2(d_im_i+d_jm_j)-4(m_i+m_j)+4}\right), \tag{8}$$

is violated by $\mathrm{DS}(k,k)$ *for all $k \geq 6$. At $k=6$,* $\mathrm{DS}(6,6)$ *with $n=14$ satisfies $\mu(G) \approx 8.772$ while the bound gives ≈ 8.676.*

Proof. The hub–leaf edge maximizes Bound 45 for $k \geq 6$. We show $\mu > B_{45}^{\mathrm{hl}}$. Both $\mu-2$ and $B_{45}^{\mathrm{hl}}-2$ are positive for $k \geq 2$, so $\mu > B_{45}^{\mathrm{hl}}$ if and only if $(\mu-2)^2 > (B_{45}^{\mathrm{hl}}-2)^2$. Expanding and simplifying, then squaring a second time to clear the remaining radical, reduces to $4k(k^3+8k^2-11k-2) > 0$, which holds for $k \geq 2$ since $k^3+8k^2-11k-2 > 0$ for $k \geq 2$. The gap satisfies $B_{45}-\mu = -1/(2k)+O(k^{-2})$. □

Proposition 4.6. *For $t \geq 2$ and $s \geq 1$, the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2)+\sqrt{(t-s)^2+4s}}{2}. \tag{9}$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \left(\begin{smallmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{smallmatrix}\right)$ with characteristic polynomial $-\lambda[\lambda^2-(t+s+2)\lambda+n]$, where $n=1+t(1+s)$. Discriminant $(t-s)^2+4s > 0$. □

Table 1: Summary of counterexamples to BHS bounds. "Gap" denotes the bound value minus $\mu(G)$ (negative = violation).

Bound	Counterexample	n	$\mu(G)$	Bound value	Gap
11	$\mathrm{SoC}(K_8, 15)$	121	16.7958	16.0000	−0.7958
13	$\mathrm{SoC}(K_9, 17)$	154	18.8151	18.0000	−0.8151
40	$\mathrm{SoS}(39, 8)$	352	40.2560	38.9818	−1.2741
45	$\mathrm{DS}(6, 6)$	14	8.7720	8.6762	−0.0958
48	P_3	3	3.0000	2.9289	−0.0711

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \tag{10}$$

is violated by $\mathrm{SoS}(39, 8)$ *with* $n = 352$*, where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982*.*

Proof. The hub–intermediate edge has $d_h = 39$, $m_h = 9$, $d_w = 9$, $m_w = 47/9$. By Proposition 4.6, $\mu = (49 + \sqrt{993})/2$. The inequality $\mu > B_{40}$ reduces to $\sqrt{993} > 27.25$, i.e., $993 > 742.6$, which holds. Similarly, $\mathrm{SoS}(49, 9)$ with $n = 491$ gives $\mu = (60 + \sqrt{1636})/2 \approx 50.224$ while $B_{40} \approx 48.487$ at the hub–intermediate edge (with $d_h = 49$, $m_h = 10$, $d_w = 10$, $m_w = 58/10$), confirming the violation with gap ≈ 1.74. □

Remark 4.8 (Bound 48 — degenerate case). BHS Bound 48,

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}},$$

is violated by P_3: the bound gives $10/(2+\sqrt{2}) \approx 2.929$ while $\mu(P_3) = 3$. This is the only counterexample found.

5. Why these bounds fail

The five bounds disproved in Section 4 fail for related but distinct structural reasons. In this section we identify two mechanisms—hub invisibility (vertex-type) and degree–spectrum decoupling (edge-type)—and show that

together they account for all five violations. We then classify all 24 vertex-type BHS bounds by their hub behavior, providing a unified framework that explains why the remaining vertex-type bounds resist our counterexample families.

5.1. Hub invisibility in vertex-type bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the hub vertex has $(d_v, m_v) = (t, m)$, the connectors have $(d_v, m_v) = \big(m,\, (t + (m-1)^2)/m\big)$, and the interior vertices have $(d_v, m_v) = \big(m-1,\, (m-2) + m/(m-1)\big)$. The Laplacian spectral radius satisfies $\mu \sim t + 1$ as $t \to \infty$ (Proposition 4.1), growing linearly in t.

Definition 5.1 (Hub-invisible bound)**.** A vertex-type bound $\max_v f(d_v, m_v)$ is *hub-invisible* on $\mathrm{SoC}(K_m, t)$ if $f(t, m) \to 0$ as $t \to \infty$ with m fixed, and *hub-visible* if $f(t, m) \to \infty$.

The following proposition classifies all 24 vertex-type BHS bounds.

Proposition 5.2 (Classification of vertex-type bounds)**.** *Among the 24 vertex-type BHS bounds (Bounds 1, 4–14, 16, 18–27, 30), exactly two are hub-invisible:*

(a) *Bound 11, $f_{11}(d, m) = 2m^3/d^2$: hub contribution $f_{11}(t, m) = 2m^3/t^2 \to 0$.*

(b) *Bound 13, $f_{13}(d, m) = 2m^4/d^3$: hub contribution $f_{13}(t, m) = 2m^4/t^3 \to 0$.*

The remaining 22 vertex-type bounds are all hub-visible: for each, $f(t, m) \to \infty$ as $t \to \infty$.

Proof. We verify the hub contribution $f(t, m)$ for each bound as $t \to \infty$ with m fixed. Representative cases:

- *Pure d-dominant bounds* (1, 4, 7, 8, 9, 14, 16): these satisfy $f(t, m) \geq c \cdot t^\alpha$ for some $\alpha \geq 1$ and $c > 0$. For example, $f_4(t, m) = 2t^2/m \to \infty$ and $f_{14}(t, m) = 2t^3/m^2 \to \infty$.

- *Mixed bounds* (5, 6, 10, 12, 18–27, 30): each contains at least one term growing in t. For example, $f_5(t, m) = t^2/m + m \to \infty$; $f_6(t, m) = \sqrt{m^2 + 3t^2} \to \infty$; $f_{30}(t, m) = m^3/t^2 + t^2/m \to \infty$ since the second term dominates.

- *Hub-invisible bounds* (11, 13): these are the only bounds where every term in f has d in the denominator with higher power than in the numerator. $f_{11}(t,m) = 2m^3/t^2 \to 0$; $f_{13}(t,m) = 2m^4/t^3 \to 0$.

The full evaluation for all 24 bounds is recorded in Table 2. □

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub vertex—which dominates the Laplacian spectrum via Proposition 4.1—becomes invisible to the bound formula. The bound is then determined by the connector contribution, which at $t = 2m - 1$ evaluates to $2m$, while $\mu > 2m$ by Proposition 4.1. Conversely, the 22 hub-visible bounds grow at least as fast as the spectral radius in t, which is why our star-of-cliques families cannot violate them.

Remark 5.3. Bounds 11 and 13 are the *only* vertex-type BHS bounds of the form $f(d,m) = 2m^p/d^{p-1}$ (with $p = 3, 4$). All other vertex-type bounds contain terms that grow with d, ensuring hub visibility. This explains the pattern observed in Section 4: the star-of-cliques family violates precisely these two bounds and no others.

5.2. Edge-type mechanisms

The three edge-type bounds we disprove (40, 45, 48) fail by mechanisms distinct from hub invisibility.

5.2.1. Bound 40: transient violation window

Bound 40 is an edge-type bound violated by $\mathrm{SoS}(t,s)$. Unlike the divergent violations of Bounds 11 and 13, Bound 40 exhibits a *transient* violation: the gap $\mu - B_{40}$ is positive only for t in a bounded interval.

Proposition 5.4 (Transient violation of Bound 40)**.** *On* $\mathrm{SoS}(t,s)$ *with* s *fixed, the violation gap of Bound 40 at the hub–intermediate edge satisfies:*

(a) *For* $t \ll s^2$*:* $B_{40} > \mu$ (bound holds).

(b) *For* $t \sim s^2$*:* $B_{40} < \mu$ (bound fails).

(c) *For* $t \gg s^2$*:* $B_{40} > \mu$ (bound holds again).

In particular, with $s = 8$ *the maximum violation gap is approximately* 1.27 *at* $t = 39$.

Proof. On $\mathrm{SoS}(t,s)$, the hub has degree $d_h = t$ and average neighbor degree $m_h = s{+}1$ (each of its t neighbors has degree $s{+}1$). Each intermediate vertex has degree $d_w = s+1$ and average neighbor degree $m_w = (t+s)/(s+1)$.

The spectral radius satisfies $\mu(\mathrm{SoS}(t,s)) = \big(t+s+2+\sqrt{(t-s)^2+4s}\big)/2$ (Proposition 4.6), which grows as $t + O(1)$ for large t.

The Bound 40 value at the hub–intermediate edge is

$$B_{40} = 2 + \sqrt{2(m_h{-}1)^2 + 2(m_w{-}1)^2 + d_h^2 + d_w^2 - d_h m_h - d_w m_w}.$$

Substituting the parameters and expanding, the dominant term under the square root for large t is $d_h^2 - d_h m_h = t^2 - t(s{+}1) = t(t{-}s{-}1)$, giving $B_{40} \sim 2 + t$ for $t \to \infty$. Since $\mu \sim t+1$, we have $B_{40} - \mu \sim 1 > 0$: the bound recovers.

For small t, both μ and B_{40} are $O(s)$, and the bound also holds. The violation occurs in a transient window near $t \approx s^2$ where the average neighbor degree term $m_w = (t{+}s)/(s{+}1)$ is large enough to inflate μ via the quotient matrix mechanism but not yet large enough to dominate the bound formula. The maximum violation gap of approximately 1.27 at $(t,s) = (39,8)$ was confirmed numerically. □

5.2.2. Bound 45: antisymmetric mode decoupling

Proposition 5.5 (Antisymmetric mode mechanism)**.** *Bound 45 fails on* $\mathrm{DS}(k,k)$ *because the Laplacian spectral radius arises from the antisymmetric eigenvector sector, where the two hub vertices carry opposite signs. The bound formula, which evaluates a symmetric expression at each edge, cannot detect this antisymmetric oscillation.*

Proof. By Proposition 4.4, the spectral radius of $\mathrm{DS}(k,k)$ is

$$\mu\big(\mathrm{DS}(k,k)\big) = \frac{k+3+\sqrt{k^2+6k+1}}{2},$$

arising from the antisymmetric ($\sigma = -1$) sector of the equitable partition quotient. The corresponding eigenvector has $x_u = -x_v$ at the two hubs, and $x_{\ell_u} = -x_{\ell_v}$ at their respective pendant leaves.

Bound 45 evaluates $g_{45}(d_i, d_j, m_i, m_j)$ at each edge. On the hub–hub edge of $\mathrm{DS}(k,k)$, we have $d_i = d_j = k+1$ and $m_i = m_j = (k + k{+}1)/(k{+}1)$. By symmetry, both edges incident to each hub yield the same value of g_{45}, yet the actual spectral radius depends on the global topology—specifically the

antisymmetric eigenmode across the hub–hub bridge. The bound formula depends only on local degree parameters and cannot distinguish the symmetric eigenvalue $\lambda_{\text{sym}} = (k+3-\sqrt{k^2+6k+1})/2$ from the antisymmetric eigenvalue $\lambda_{\text{anti}} = (k+3+\sqrt{k^2+6k+1})/2$.

The violation gap satisfies $\mu - B_{45} = \Theta(1/k)$ as $k \to \infty$, so the gap vanishes asymptotically. This suggests the bound is *repairable* with an additive $O(1/\Delta)$ correction term. □

5.2.3. Bound 48: degenerate small-graph failure

Bound 48 is violated only by P_3. Unlike the other four violations, this appears to be a degenerate case: the bound formula contains a denominator $2+\sqrt{\cdots}$ that assumes nontrivial local structure. On P_3, with $d_i = 1$, $d_j = 2$ at the pendant–center edge, the discriminant collapses and the bound evaluates to $10/(2+\sqrt{2}) < 3 = \mu(P_3)$. No counterexample was found for $n \geq 4$ among all graphs tested (Section 6.3), and we conjecture that the bound holds for all connected graphs with $n \geq 4$.

5.3. Unified mechanism summary

Table 3 summarizes the failure mechanism for each disproved bound.

The two vertex-type failures (Bounds 11, 13) are *irreparable*: the violation gap diverges, so no bounded correction term can salvage these bounds. Bound 45 is asymptotically repairable since the gap vanishes as $O(1/\Delta)$. Bound 48 is trivially repairable by excluding P_3. Bound 40 occupies an intermediate position: the violation is bounded but occurs over a range of parameters, making a simple correction nontrivial.

6. Status of the 38 open BHS bounds

Of the 68 bounds proposed by BHS [2], Ghebleh et al. [3] disproved 30, leaving 38 open. We have disproved 5 of these 38 (Section 4), leaving **33 bounds open**.

6.1. Classification criteria

We classify the 33 remaining open bounds into four categories based on their behavior across all counterexample families tested ($\text{SoC}(K_m, t)$ for $m \in [3, 12]$, $t \in [2, 200]$; $\text{SoS}(t, s)$ for $t \in [2, 100]$, $s \in [1, 20]$; $\text{DS}(k, k)$ for $k \in [1, 50]$; $\text{Cat}(\ell, k)$ for $\ell \in [2, 10]$, $k \in [0, 50]$; and all connected graphs with $n \leq 10$):

Definition 6.1 (Classification criteria)**.** Let $\mathrm{gap}(B, G) = B(G) - \mu(G)$ denote the margin by which bound B holds on graph G (positive means the bound holds).

(i) **Super-safe**: The relative gap $\mathrm{gap}(B, G)/\mu(G)$ exceeds 0.1 on all tested families (large relative margin).

(ii) **Tight**: The infimum of $\mathrm{gap}(B, G)$ over all tested families equals zero (exact equality achieved on some graph family).

(iii) **Safe**: The infimum of $\mathrm{gap}(B, G)/\mu(G)$ over all tested families is bounded below by a positive constant (relative gap bounded away from zero).

(iv) **Marginal**: Neither tight nor safe—the absolute gap is positive on all tested graphs but the relative gap approaches zero along some family.

6.2. Complete status table

Table 4 presents the complete status of all 38 bounds that were open after [3], organized by resolution status.
Of the 33 remaining open bounds, all vertex-type ones are hub-visible (Proposition 5.2). Three edge-type bounds (33, 42, and 47) are *tight*: they achieve exact equality on complete bipartite graphs and windmill graphs.

Remark 6.2 (Comparison with Ghebleh et al.). Of the 30 bounds disproved by Ghebleh et al. [3], all counterexamples have at most 13 vertices. None of the 38 bounds left open by their search can be disproved by any graph with $n \leq 13$, and our smallest new counterexample (P_3 aside, which has $n = 3$) has $n = 14$ (for Bound 45). The largest has $n = 352$ (for Bound 40). This suggests a phase transition in counterexample difficulty: the "easy" bounds are disproved by small graphs findable by stochastic search, while the "hard" ones require structured infinite families with closed-form spectral analysis.

6.3. Exhaustive verification of the tight bounds

We performed exhaustive computational verification of the three tight bounds (33, 42, and 47) on every connected graph with $n \leq 10$ vertices—a total of 11,988,768 graphs. Zero violations were found.

Proposition 6.3 (Finite verification)**.** *Bounds 33, 42, and 47 from [2] hold for all connected graphs with at most* 10 *vertices.*

Windmill graphs $W(k,3)$ (the friendship graphs) provide asymptotically tight instances for Bounds 42 and 47: the relative gap satisfies $\big(B_{42}-\mu\big)/\mu \sim \sqrt{3}/k \to 0$ as $k \to \infty$. Proving or disproving these three bounds remains the most compelling open problem arising from this work.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [3] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction. Together, 35 of the original 68 bounds are now resolved (52%), with 33 remaining open.

Structural vs. computational approaches.. Cross-entropy search [3, 4, 5] explores adjacency matrices for $n \leq 24$. Our structural approach targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary: stochastic search excels at finding small counterexamples, while structural construction reaches the large-n regime ($n = 352$ for Bound 40) where no counterexample can be found computationally.

Gap asymptotics and repairability.. The five violations exhibit qualitatively different gap behaviors (Table 3). Bounds 11 and 13 have divergent gaps (irreparable). Bound 40 has a bounded transient window (max gap ≈ 1.27). Bound 45 has a vanishing gap $\Theta(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction. Bound 48 fails only on P_3. Understanding which bounds are repairable may guide the design of improved automated conjectures.

A weighted Laplacian intermediate result.. In attempting to prove the tight bounds (42, 47), we obtained the following intermediate result, which we include here as a potentially useful tool for future work.

Proposition 7.1 (Weighted Laplacian bound)**.** *For any connected graph G,*

$$\mu(G)^2 \leq \lambda_{\max}(M),$$

where M is the Laplacian of the weighted graph on the same vertex set with edge weights $w_{ij} = d_i + d_j$ for each $\{i,j\} \in E$.

Proof sketch. Let x be the eigenvector of $L(G)$ with eigenvalue $\mu(G)$, normalized so that $\|x\|_2 = 1$. The eigenequation gives $\mu(G)\, x_v = d_v x_v - \sum_{u\sim v} x_u$. Squaring both sides:

$$\mu(G)^2\, x_v^2 = \Big(d_v x_v - \sum_{u\sim v} x_u\Big)^2.$$

Expanding the square gives $d_v^2 x_v^2 - 2d_v x_v \sum_{u\sim v} x_u + \big(\sum_{u\sim v} x_u\big)^2$. Applying Cauchy–Schwarz to the cross-term, $-2d_v x_v \sum_{u\sim v} x_u \le d_v \sum_{u\sim v}(x_v^2 + x_u^2)$, and to $\big(\sum_{u\sim v} x_u\big)^2 \le d_v \sum_{u\sim v} x_u^2$, then summing over all vertices yields $\mu(G)^2 \le x^T M\, x$, where M has entries

$$M_{vv} = \sum_{u\sim v}(d_v + d_u), \qquad M_{vu} = -(d_v + d_u) \ \text{ if } u \sim v.$$

The Rayleigh quotient bound gives $x^T M\, x \le \lambda_{\max}(M)$. □

Remark 7.2. Proposition 7.1 implies that proving Bound 42, for example, reduces to showing $\lambda_{\max}(M) \le \max_{(i,j)\in E}(d_i^2 + d_j^2 + 2m_i \cdot m_j)$. This second inequality remains open.

Proof attempts for tight bounds.. We attempted to prove the two remaining tight bounds (42, 47) using Rayleigh quotient methods, Schur complement analysis, Gershgorin circles, and degree sequence constraints. While the weighted Laplacian bound (Proposition 7.1) provides a useful intermediate step, complete proofs remain elusive. The fundamental obstacle is that standard spectral techniques do not naturally produce edge-maximum bounds involving average neighbor degrees: the Rayleigh quotient $x^T Lx/x^T x$ yields vertex-sum expressions, whereas edge-type bounds require converting these into edge-maximum form.

Open questions..

(a) Can the three tight bounds (33, 42, 47) be proved?

(b) Can additional bounds among the remaining 33 be proved?

(c) Do three-level hierarchies (e.g., star-of-stars-of-cliques) violate additional bounds?

(d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

(e) What is the correct minimum-order threshold for Bound 48?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4.6) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

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Table 2: Hub behavior of all 24 vertex-type BHS bounds on $\mathrm{SoC}(K_m, t)$ as $t \to \infty$ with m fixed. Hub contribution $f(t, m)$ and its limit determine whether the bound is hub-visible (V) or hub-invisible (I). Status: DISPROVED = disproved in Section 4; OPEN = unresolved.

Bound	Formula $f(d, m)$	$f(t, m)$	Hub	Status
1	$\sqrt{4d^3/m}$	$\sim 2t^{3/2}/\sqrt{m}$	V	Open
4	$2d^2/m$	$2t^2/m$	V	Open
5	$d^2/m + m$	$t^2/m + m$	V	Open
6	$\sqrt{m^2 + 3d^2}$	$\sqrt{3}\,t$	V	Open
7	$d^2/m + d$	$t^2/m + t$	V	Open
8	$\sqrt{d(m + 3d)}$	$\sqrt{3}\,t$	V	Open
9	$(m + 3d)/2$	$3t/2$	V	Open
10	$\sqrt{d(d + 3m)}$	t	V	Open
11	$2m^3/d^2$	$2m^3/t^2 \to 0$	**I**	**Disproved**
12	$\sqrt{2m^2 + 2d^2}$	$\sqrt{2}\,t$	V	Open
13	$2m^4/d^3$	$2m^4/t^3 \to 0$	**I**	**Disproved**
14	$2d^3/m^2$	$2t^3/m^2$	V	Open
16	$2d^4/m^3$	$2t^4/m^3$	V	Open
18	$\sqrt{2m^3/d + 2d^2}$	$\sqrt{2}\,t$	V	Open
19	$(4d^4+12dm^3)^{1/4}$	$\sqrt{2}\,t$	V	Open
20	$\frac{1}{2}\sqrt{7d^2+9m^2}$	$\sqrt{7}\,t/2$	V	Open
21	$\sqrt{d^3/m + 3m^2}$	$t^{3/2}/\sqrt{m}$	V	Open
22	$(2d^4+14d^2m^2)^{1/4}$	$\sqrt[4]{2}\,t$	V	Open
23	$\sqrt{d^2 + 3dm}$	t	V	Open
24	$(6d^4+10m^4)^{1/4}$	$\sqrt[4]{6}\,t$	V	Open
25	$(3d^4+13d^2m^2)^{1/4}$	$\sqrt[4]{3}\,t$	V	Open
26	$\frac{1}{2}\sqrt{5d^2+11dm}$	$\sqrt{5}\,t/2$	V	Open
27	$\sqrt{(3d^2+5dm)/2}$	$\sqrt{3/2}\,t$	V	Open
30	$m^3/d^2 + d^2/m$	t^2/m	V	Open

Table 3: Failure mechanisms for the five disproved BHS bounds. "Gap behavior" describes $\mu(G) - B(G)$ as graph parameters grow. "Repairable" indicates whether an additive correction could restore validity.

Bound	Type	Mechanism	Repairable?	Gap behavior
11	Vertex	Hub-invisible	No	$\to +\infty$
13	Vertex	Hub-invisible	No	$\to +\infty$
40	Edge	Transient window	Unclear	Bounded (max ≈ 1.27)
45	Edge	Antisym. decoupling	Yes	$\to 0$ as $\Theta(1/k)$
48	Edge	Degenerate (P_3)	Yes	Fixed (0.071)

Table 4: Status of the 38 BHS bounds left open by Ghebleh et al. [3]. Bounds are grouped by resolution status. The 33 bounds that remain open are classified as Super-safe, Tight, Safe, or Marginal per Definition 6.1.

Bound	Type	Resolved by	Method	Category
Disproved (5 bounds) — this paper				
11	Vertex	This paper	$\mathrm{SoC}(K_m, t)$	Hub-invisible
13	Vertex	This paper	$\mathrm{SoC}(K_m, t)$	Hub-invisible
40	Edge	This paper	$\mathrm{SoS}(t, s)$	Transient window
45	Edge	This paper	$\mathrm{DS}(k, k)$	Antisymmetric mode
48	Edge	This paper	P_3	Degenerate
Still open (33 bounds)				
1	Vertex	—	—	Super-safe
4	Vertex	—	—	Super-safe
5	Vertex	—	—	Super-safe
6	Vertex	—	—	Super-safe
7	Vertex	—	—	Super-safe
8	Vertex	—	—	Safe
9	Vertex	—	—	Super-safe
10	Vertex	—	—	Marginal
12	Vertex	—	—	Super-safe
14	Vertex	—	—	Super-safe
16	Vertex	—	—	Super-safe
18	Vertex	—	—	Safe
19	Vertex	—	—	Safe
20	Vertex	—	—	Safe
21	Vertex	—	—	Safe
22	Vertex	—	—	Safe
23	Vertex	—	—	Marginal
24	Vertex	—	—	Safe
25	Vertex	—	—	Safe
26	Vertex	—	—	Safe
27	Vertex	—	—	Safe
30	Vertex	—	—	Safe
33	Edge	—	—	Tight
34	Edge	—	—	Safe
35	Edge	—	—	Safe
37	Edge	—	—	Safe
38	Edge	—	—	Safe
39	Edge	—	—	Safe
42	Edge	—	—	Tight
44	Edge	—	—	Safe
46	Edge	—	—	Safe
47	Edge	—	—	Tight
56	Edge	—	—	Safe